

Fundamentals of AI and ML

(CSA2001)

Project Report: AI Travel Planner

Name: Aklesh Swain

Registration No.: 25BAI11184

Branch: Computer Science & Engineering

Specialisation: Artificial Intelligence and Machine Learning

Year: 1st Year

Project Abstract

The **AI Travel Planner** is a sophisticated desktop application designed to simplify the complexity of travel itinerary creation. By leveraging a structured knowledge base of global heritage sites and cultural landmarks, the application generates comprehensive, time-blocked schedules. It provides users with a seamless transition from digital planning to a professional-grade physical or digital PDF document, ensuring all logistical details from morning excursions to evening relaxation are curated for an optimal travel experience.

Technical Specifications

- **User Interface:** A custom-themed Dark Mode interface built with Tkinter, featuring a gold-and-slate aesthetic for a premium user experience.
- **Document Generation:** Integration of the ReportLab library to programmatically generate A4-sized PDF itineraries with custom branding, table styles, and color-coded headers.
- **Concurrency Management:** Implementation of threading to handle intensive PDF generation and "AI simulation" steps (like route optimization and heritage mapping) without interrupting the UI responsiveness.
- **Data Architecture:** A multi-layered dictionary system that stores historical context, activity descriptions, and time-specific tags for over a dozen major cultural hubs.
- **Intelligent Recommendation Engine (KNN):** Implementation of a **K-Nearest Neighbours (KNN)** algorithm using scikit-learn. The system performs multi-dimensional vector matching between user preferences and a city-interest matrix. It utilizes **MinMaxScaler** for feature normalization and **Cosine Similarity** to ensure high-precision destination matching based on unique cultural and travel attributes.
- **Natural Language Processing (NLP) Module:** A custom-built, rule-based **NLP Engine** that utilizes regular expressions (re) and lexicon mapping. The engine

performs entity extraction to identify city names and intent classification to map user-typed keywords.

AI-Powered Travel Planner

Plan Your Journey

Tell us where you want to go and our AI will craft a personalised day-by-day heritage travel itinerary.

DESTINATION

-- Select a city --

NLP: describe your trip (optional)

Enter place Name

DURATION (DAYS)

235710

BUDGET

\$

Budget

< Rs.5k/day

**

Mid

Rs.5-15k/day

Crown

Luxury

Rs.25k+ /day

INTERESTS

HistoryArchitectureReligionNatureFood

PhotographyCultureAdventureShoppingNightlife

Map

Your AI Itinerary Awaits

Fill in your travel preferences and let AI craft your perfect India heritage journey.

- Choose a destination with rich monument history
- Select interests tailored to your preferences
- Longer durations unlock hidden gems nearby
- Budget affects hotel and dining suggestions

6 PM
Moon

times after destruction; the current golden spire structure was commissioned by Maratha queen Ahilyabai Holkar in 1780. The corridor was newly expanded in 2021. Entry: Free.

TempleHinduHistory

6 PM
Moon

EVENING

Ganga Aarti at Dashashwamedh Ghat

The grandest of Varanasi's 84 ghats, Dashashwamedh is where Lord Brahma is said to have performed the Dasa Ashwamedh yagna (10-horse sacrifice). Every evening at dusk, a team of priests simultaneously perform a choreographed fire ritual with large brass diyas, conch shells, and Sanskrit chanting — drawing thousands of devotees and visitors.

RitualFireSacred

D2

Day 2 - Sarnath & the Old City

Sun, 05 Apr

D3

Day 3 - The 84 Ghats — A Complete Walk

Mon, 06 Apr

Download PDF

Share Link

Save

Key Features

- **Intelligent Itinerary Logic:** Users can define their destination, trip duration (up to 10 days), and travel pace. The system dynamically scales the content to fit the requested timeframe.
- **Budget Tiering:** Integrated estimation for **Budget**, **Mid-range**, and **Luxury** travel styles, helping users plan according to financial preferences.
- **Time-Block Scheduling:** Each day is divided into three distinct phases:
 - **Morning (7:00 AM):** High-energy heritage visits.
 - **Afternoon (1:00 PM):** Cultural immersion and museums.
 - **Evening (6:00 PM):** Local cuisine and leisure.
- **Professional Export Options:** * **PDF Report:** A formal document with embedded metadata and structured tables.
 - **Digital Share:** Generation of a temporary HTML/browser-viewable version for quick mobile access.

Walk Through History

Varanasi, Uttar Pradesh Heritage Journey

The Eternal City on the Ganges

3 Days

Mid Budget

Relaxed Pace

3

9

6

Rs.7,500

DAYS

EXPERIENCES

MONUMENTS

EST./DAY

D1

Day 1 - The Ghats & Sacred Fire

Saturday, 04 Apr 2026

7 AM

MORNING - 7 AM

Sunrise Boat Ride on the Ganges

Varanasi's 84 ghats stretch for 6.5km along the western bank of the Ganges. A sunrise boat ride (cost -Rs.300-500) reveals the full spectacle of daily Hindu life — priests, pilgrims bathing, yoga practitioners, and the burning pyres of Manikarnika Ghat. This is considered one of the world's oldest continuously inhabited cities.

Sunset | Photography | Culture

1 PM

AFTERNOON - 1 PM

Kashi Vishwanath Temple

One of the 12 Jyotirlingas (sacred abodes of Lord Shiva) and among the most important temples in Hinduism. The temple has been rebuilt multiple times after destruction; the current golden spire structure was commissioned by Maratha queen Jhansi Bai in 1780. The corridor was newly expanded in 2021. Entry: Free.

Temple | History | History

6 PM

EVENING - 6 PM

Ganga Aarti at Dashashwamedh Ghat

The grandest of Varanasi's 84 ghats, Dashashwamedh is where Lord Brahma is said to have performed the Dashashwamedh yagna (10 horse sacrifice). Every evening at dusk, a team of priests simultaneously perform a choreographed fire ritual with large brass diya, conch shells, and Sanskrit chanting — drawing thousands of devotees and visitors.

Ritual | Fire | Sacred

D2

Day 2 - Sarnath & the Old City

Sunday, 05 Apr 2026

7 AM

MORNING - 7 AM

Sarnath — The Deer Park

Located 10km north of Varanasi, Sarnath is one of the four most sacred Buddhist pilgrimage sites. It was here, in the Vapasia Deer Park, that the Buddha delivered his first sermon (the Dhammacakkapavattana Sutta) in 5th century BCE. The Dhamek Stupa (500 CE) marks the exact spot. Entry: Rs.25 (Indians).
Buddhist | Dharmic | History

AFTERNOON - 1 PM

Ramnagar Fort & Museum

Built in the 18th century by Maharaja Balwant Singh, this cream-coloured sandstone fort across the Ganges is the ancestral seat of the Maharaja of Varanasi. The Veda Vyas Museum inside houses a remarkable collection of medieval weapons, royal palanquins, vintage automobiles, and rare astronomical instruments.

Fort | Royalty | Museum

6 PM

EVENING - 6 PM

Banarasi Silk Weaving Quarter

Varanasi's silk weaving tradition dates back over 2,000 years. The city's weavers — concentrated in the Peshi Kuthi and Wadapur neighbourhoods — produce the celebrated Banarasi silk saris, known for their gold and silver brocade (zari) work. A guided workshop visit reveals the intricate handloom process. Cost: Free-Rs.200.

Craft | Culture | Shopping

D3

Day 3 - The 84 Ghats — A Complete Walk

Monday, 06 Apr 2026

7 AM

MORNING - 7 AM

Manikarnika Ghat — The Burning Ghat

Manikarnika is one of the most sacred cremation sites in Hinduism — Hindus believe dying here guarantees moksha (liberation from the cycle of rebirth). Fires have burned here continuously for 3,500 years according to tradition. The ghat is managed by the Dom community, hereditary keepers of the cremation fires. Approach with deep respect and no cameras. Entry: Free.

Funeral | Hindu | Culture

1 PM

AFTERNOON - 1 PM

Asst to Raj Ghat — Full Ghat Walk

Walk the 6.5km stretch of all 84 ghats from Assi Ghat (south) to Raj Ghat (north) — an extraordinary 3-hour journey through every dimension of Varanasi life. Each ghat has a distinct character: Harishchandra (a sacred burning ghat), Kedar Ghat (South Indian pilgrims), Darbhanga Ghat (a palace), and Panchsengha Ghat (five sacred rivers).

Walking | Culture | Heritage

6 PM

EVENING - 6 PM

Evening Aarti at Multiple Ghats

Beyond Dashashwamedh, smaller aartis happen simultaneously at Assi Ghat (more intimate, popular with locals), Rajendra Prasad Ghat, and Panchsengha Ghat. Witnessing the same ancient ritual at three different locations on the same evening gives a sense of the city's layered spiritual geography. Cost: Free.

Ritual | Sacred | Photography

Generated by Walk Through History - 28 Mar 2026, 16:54

* AI-Powered Travel Planner

Plan Your Journey

Tell us where you want to go and our AI will craft a personalised day-by-day heritage travel itinerary.

BUDGET

\$

Budget

< Rs.5k/day

**

Mid

Rs.5-15k/day

Crown

Luxury

Rs.25k+/day

INTERESTS

History

Architecture

Religion

Nature

Food

Photography

Culture

Adventure

Shopping

Nightlife

TRAVEL STYLE

Walk

Relaxed

Fast

Packed

Focus

Focused

Free

Flexible

TRAVELLERS

Solo

Couple

Family

Group

Generating...

Crafting your itinerary...

Crafting morning activities...

C:/Users/Amarnath%k20Swain/AppData/Local/Temp/travel_itinerary_6ewc11om.html

* Heritage Itinerary

Varanasi, Uttar Pradesh Heritage Journey

The Eternal City on the Ganges

2 Days

Mid Budget

Relaxed Pace

2

6

6

Rs.7,500

DAYS

EXPERIENCES

MONUMENTS

EST./DAY

D1

Day 1 — The Ghats & Sacred Fire

Saturday, 04 Apr 2026

7 AM

Sunrise

MORNING

Sunrise Boat Ride on the Ganges

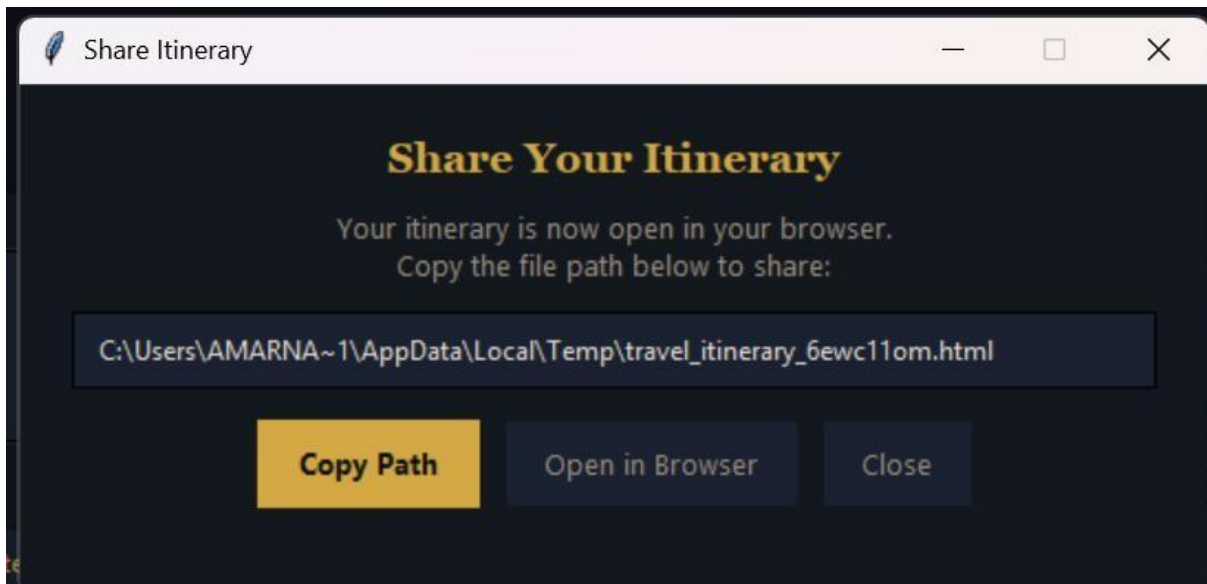
Varanasi's 84 ghats stretch for 6.5km along the western bank of the Ganges. A sunrise boat ride (cost -Rs.300-500) reveals the full spectacle of daily Hindu life — priests, pilgrims bathing, yoga practitioners, and the burning pyres of Manikarnika Ghat. This is considered one of the world's oldest continuously inhabited cities.

Sunset | Photography | Culture

1 PM

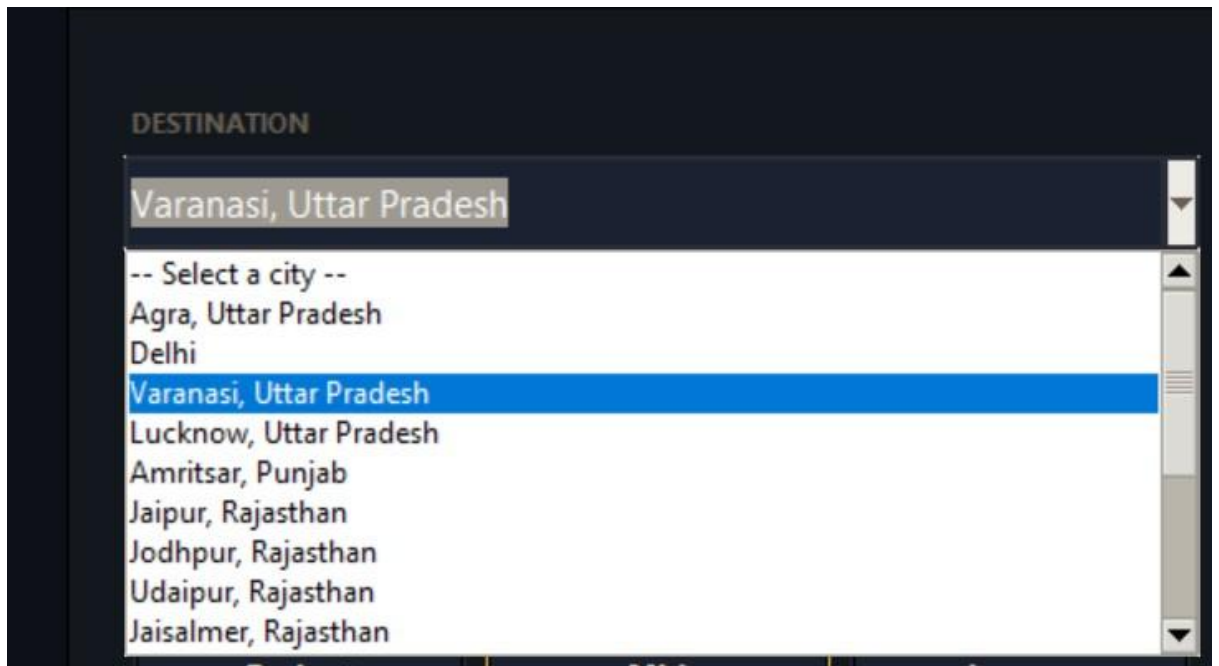
Sun

AFTERNOON



System Workflow

1. **Input Phase:** The user selects a destination (e.g., Agra, Delhi, Jaipur) and configures trip parameters via the dashboard.
2. **Simulation Phase:** The "AI Engine" performs a series of mock-processing steps, including mapping heritage sites by era and optimizing travel distances.
3. **Assembly Phase:** The SimpleDocTemplate engine gathers the data, applies Paragraph Styles, and constructs the visual layout.
4. **Output Phase:** The final report is saved to the local system and can be instantly opened for review or printing.



Future Enhancements

- **Real-time API Integration:** Connecting to live weather and Google Maps APIs for real-time travel alerts.
- **Machine Learning Integration:** Analysing user feedback to refine destination suggestions based on past preferences.
- **Multi-Language Support:** Localizing the generated reports for international travellers.

Conclusion

The AI Travel Planner successfully bridges the gap between automated data processing and usercentric design, providing a high-utility tool for modern travellers seeking structured exploration.

Thank You